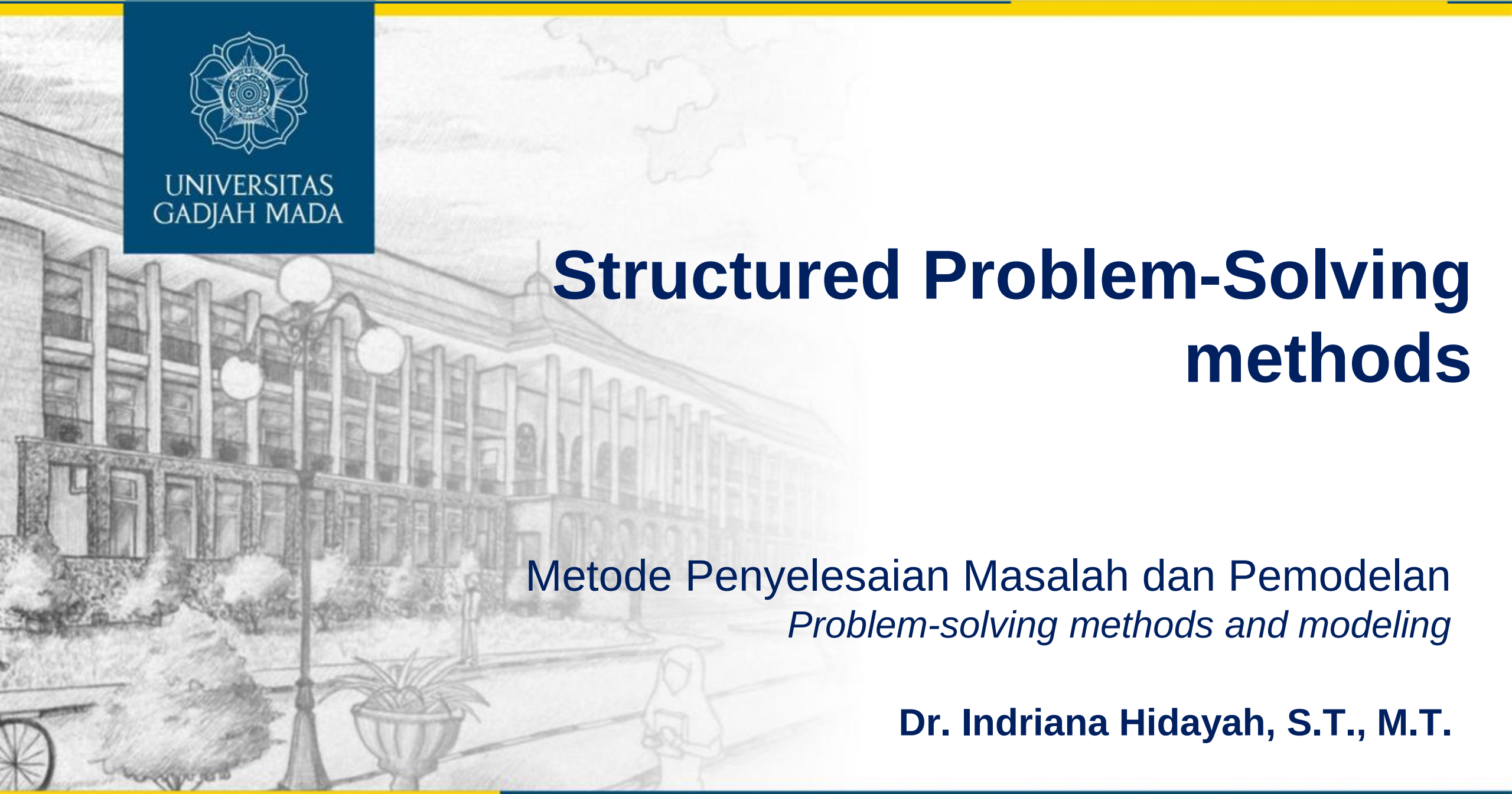


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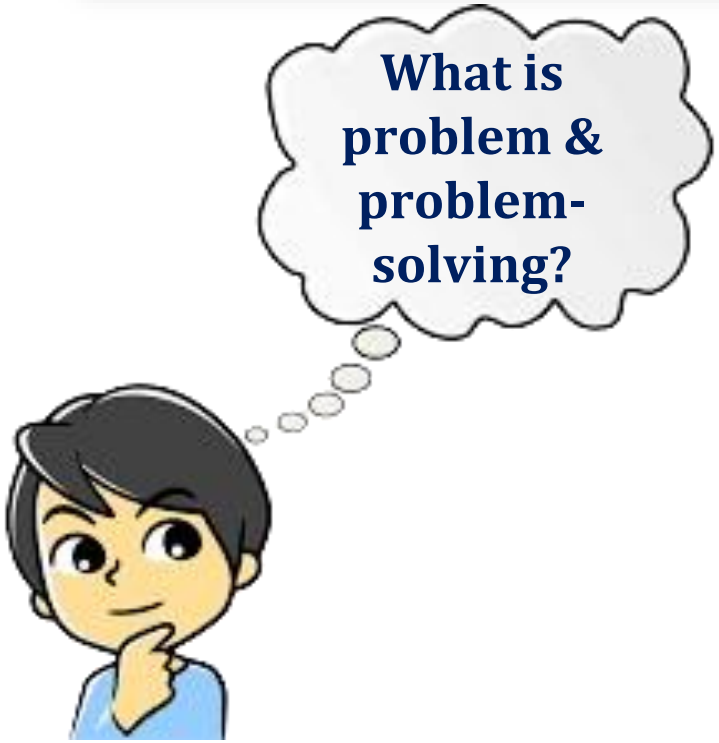
Structured Problem-Solving methods

Metode Penyelesaian Masalah dan Pemodelan

Problem-solving methods and modeling

Dr. Indriana Hidayah, S.T., M.T.

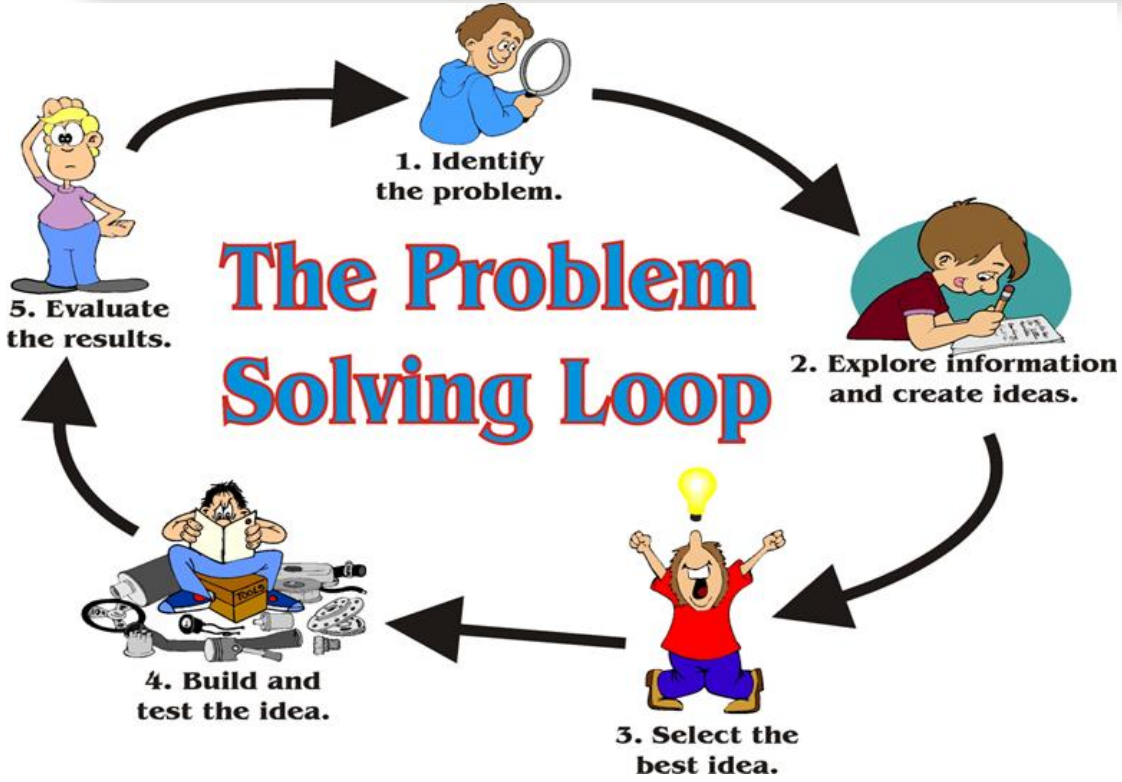
Problem-Solving



A **problem** is a situation, question, or obstacle that creates a gap between the current state and a desired outcome, requiring a solution or resolution. Problems arise when there is a discrepancy between expectations and reality, preventing progress or achieving goals.

Problem-solving is is the process of identifying a challenge or issue or problem and finding a solution.

General problem-solving steps



- One goal of this course is to train your problem-solving skill.

General problem-solving steps

- Identifying the Problem: Recognize and define the problem clearly.
- Gathering Information: Collect relevant data and information about the problem.
- Generating Solutions: Brainstorm possible solutions or approaches to address the problem.
- Evaluating Solutions: Assess the potential solutions and weigh their pros and cons.
- Choosing a Solution: Select the most appropriate and feasible solution.
- Implementing the Solution: Put the chosen solution into action.
- Reviewing the Results: Evaluate the outcome to see if the problem has been resolved effectively.

Identify problem: example

- Problem Statement: The study period of graduate students is often longer than expected.
  - **Clarify the problem**
    - **A tool called "is/is not diagram" can help in focusing on the true nature of the problem by eliminating assumptions and irrelevant details.**

Problem statement	Details
What it IS	Students need longer time to complete their study.
	…
What it IS NOT	Not exclusively cause by poor academic performance.
	…

Example

What it IS	Details
What it IS	- Graduate students taking longer than the expected duration to complete their studies.
	- Common across multiple departments and programs.
	- Occurs despite initial planning and timelines.
	- Leads to delays in graduation and entry into the job market.
	- Affects students' financial and personal plans.
What it IS NOT	- Not limited to a specific group of students (e.g., part-time vs. full-time).
	- Not exclusively caused by poor academic performance or failure.
	- Not a recent trend; it has been an ongoing issue.
	- Not unique to one university or institution.
	- Not related to the availability of resources (e.g., libraries, labs).

Identify problem: example

- Problem Statement: The study period of graduate students is often longer than expected.
  - **Possible Contributing Factors to Analyze**
    - **"is/is not diagram" can also be used to help in this task.**

Possible contributing factors	IS a contributing factor	IS NOT a contributing factor
Academic workload	X	
Enrollment in required courses		X
…		

Example

- Problem Statement: The study period of graduate students is often longer than expected.
  - **Possible Contributing Factors to Analyze**
    - **What it IS:**
      - Time Management: Graduate students may struggle with managing their time effectively.
      - Research Complexity: The complexity of research projects may lead to extended study periods.
      - Advisor Support: Variability in advisor availability and support can impact progress.
      - Funding: Limited or fluctuating funding can cause delays.
      - Life Events: Personal life events (e.g., family responsibilities, health issues) can extend study periods.

Example

- Problem Statement: The study period of graduate students is often longer than expected.
  - **Possible Contributing Factors to Analyze**
    - **What it IS NOT:**
      - Academic Failure: The delays are not primarily due to academic failures or poor grades.
      - Resource Availability: It is not due to the lack of access to necessary resources like libraries or laboratories.
      - Limited to One Institution: This is not a problem unique to a single institution; it occurs across multiple universities.

Next Steps...?

- **Gather Data:**
  - **Collect data on time management practices,**
  - **research complexities,**
  - **advisor support,**
  - **funding stability, and**
  - **personal life events -> mental health**

Next Steps...?

- **Develop solution:**
  - **Generating Solutions: Brainstorm possible solutions**
  - **Evaluating Solutions: Assess the potential solutions and weigh their pros and cons.**
  - **Choosing a Solution: Select the most appropriate and feasible solution.**
- **Implementation**
- **Review the result**

Some structured Problem-Solving Methods

To be able to perform problem-solving, there are several methods that can be used, including:



- PDCA Cycle (Plan Do Check Act)
- Root Cause Analysis
- Define Measure Analyze Improve Control (DMAIC)
- 5 Whys Technique
- Theory of Inventive Problem Solving (TRIZ)

PDCA Cycle (Plan Do Check Act)

Definition

The PDCA cycle (Plan Do Check Act) is a methodology to develop and customize internal organizational procedures for quality, project, and problem management

The PDCA cycle (Plan Do Check Act) comprises four iterative steps:

- Plan (set objectives and develop a plan to meet a need)
- Do (implement the plan)
- Check (collect data and evaluate results)
- Act (take corrective action and standardize processes)





Step 1: Plan (PDCA)

The first step in the PDCA cycle (Plan) has four sub-steps:

1. **Identify problems**: The aim is to track down the root causes of problems.
   Example → If customers complain about product quality, it's crucial to identify the factors underlying the problem to eliminate it for good.

2. **Set objectives**: The organization's overall targets should serve as a focus for short- and medium-term objectives.
   Example → A manufacturing company might set itself the target of reducing product defects by 20% over the next six months. However, these objectives need to be SMART (specific, measurable, achievable, realistic, and timely) to see conclusive results.

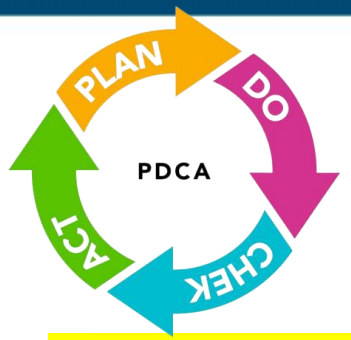


Step 1: Plan (PDCA)

3 **Develop a detailed plan:** Specify what needs to be accomplished, who is responsible for each task, how the implementation schedule will be phased, and what resources are required.
Example → A construction company looking to improve efficiency on its job sites, the detailed plan could include:

- Creating Gantt charts for each project
- Planning crew assignments
- Managing resources
- Purchasing new technologies to speed up construction processes

4 **Communicate the plan**: The final sub-step of the planning phase is sharing this plan with the whole organization. Everyone needs to understand their role and responsibilities in executing the PDCA framework before its implementation.



Step 2: Do (PDCA)

The Do phase of the PDCA process is used to transform planning into concrete action with solutions to the initial problem. It is made up of five sub-phases.

1. **Allocate the necessary resources:** Regarding workforce, materials, equipment, or budget, this ensures that the teams involved have the right resources to meet deadlines and quality standards.

2. **Train employees:** Training enables staff to acquire the skills to adapt to change, effectively use new tools, and contribute to successful implementation.

3. **Implement the plan:** Carry out the actions or tasks described methodically. For example, when introducing a new process in the manufacturing sector, it is essential to implement the plan and rigorously follow the steps to minimize potential losses.



Step 2: Do (PDCA)

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4 **Meet Deadlines:** It's imperative to meet the deadlines and milestones defined in the project plan to achieve the desired results. This discipline ensures that the project progresses according to schedule and avoids costly delays.

5 **Document:** Record all procedures and protocol details to meet regulatory requirements and enable in-depth performance analysis (equipment and materials used, unexpected problems, and deviations from the plan).



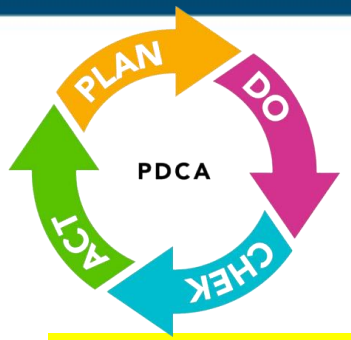
Step 3: Check (PDCA)

The Check step in the PDCA cycle is crucial for assessing the effectiveness of the actions taken. Gaps in organizational processes often become apparent at this stage. Therefore, it is essential to incorporate this step from the initial planning phase. The sub-steps include:

1. **Collect data:** Evaluate the effectiveness of the measures taken in the Do phase.
   Example: A manufacturing environment, collecting data on machine performance, downtime, and product quality is essential for assessing the efficiency of operations.

2. **Compare with predefined objectives**: Determining whether the plan has achieved the desired results is essential.
   Example → The distribution sector, comparing actual stock levels with the numbers defined at the planning stage helps to determine whether operations are on target.

3. **Analyze the root cause:** This is necessary in the event of deviation from objectives. When nonconformance to a clinical protocol occurs in the healthcare sector, it is essential to analyze the root causes, such as inadequate medical staff training or resource constraints, to ensure that patients receive quality care.



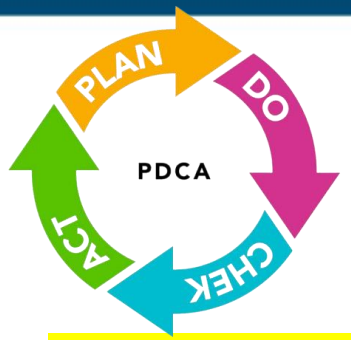
Step 3: Check (PDCA)

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4. **Share results:** Sharing results with all workers and managers promotes a culture of transparency and safety.

5. **Maintain or revisit the plan:** After assessing the performance of a strategy, you need to decide whether to keep, adjust, or review the approach if the results don't meet expectations.

6. **Document:** Future progress and permanent problem resolution depend on this sub-step. For example, in the healthcare field, detailed documentation of patient medical records and clinical procedures is necessary to ensure complete healthcare traceability and meet regulatory requirements.

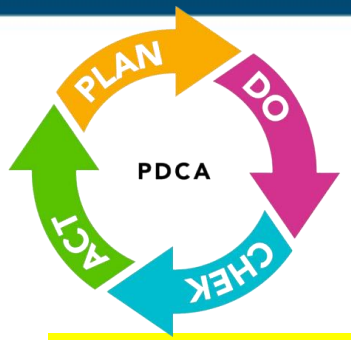


Step 4: Act (PDCA)

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1. **Adjust and reassess:** When results fall short of expectations, adjustments and reassessment are necessary.
   Example → If a company fails to meet its target of reducing product defects by 20% in six months, it can review its initial plan and make further changes.

2. **Provide feedback:** Feedback channels play a vital role in this phase. They enable the gathering of input from employees, customers, and other stakeholders.
   Example → Customer comments on product quality may indicate the need for further improvements.



Step 4: Act (PDCA)

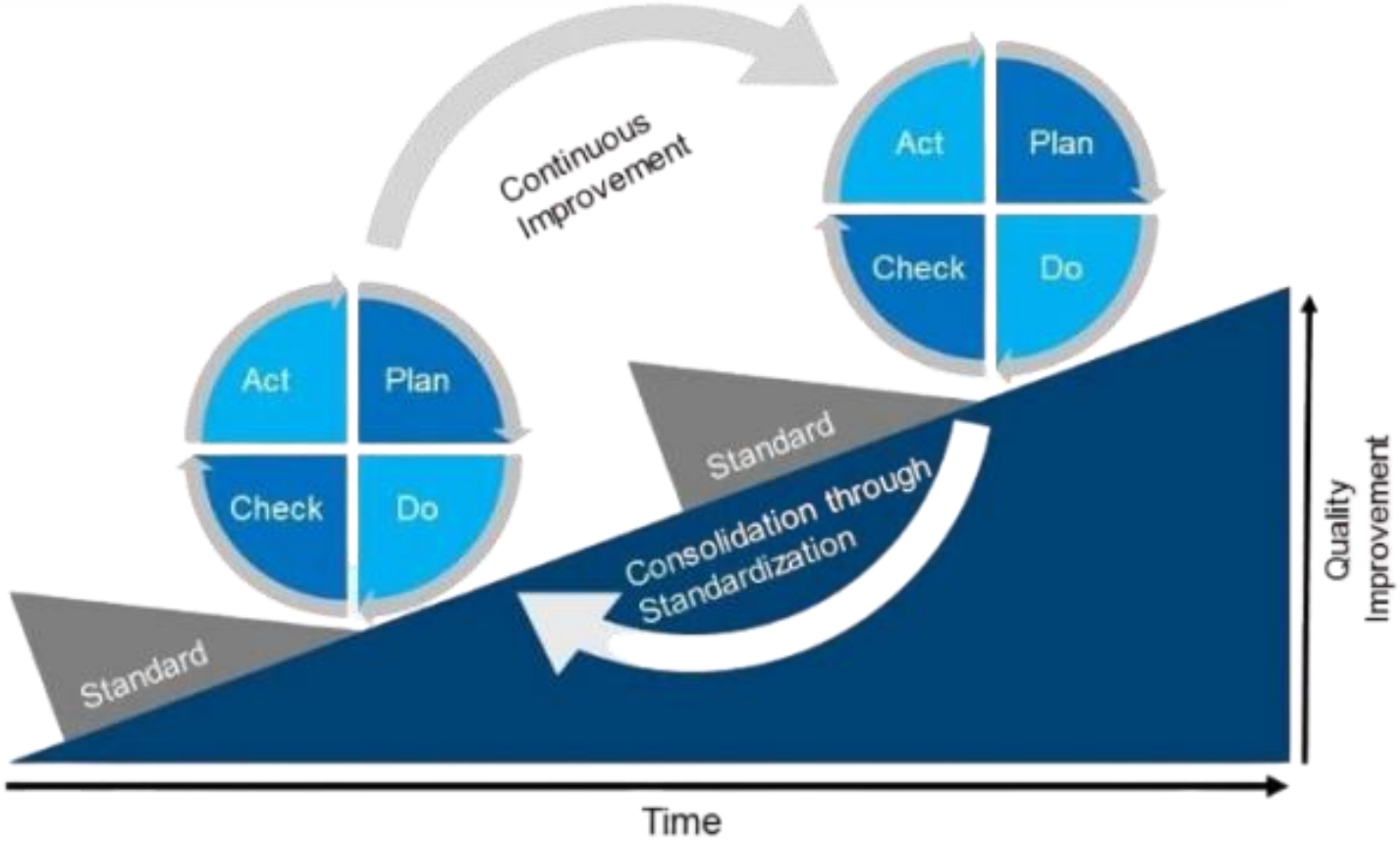
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3 **Standardize processes:** As part of the Act step, it is essential to deploy corrective measures and standardize all procedures. Standardization ensures that improvements are systematically integrated, and activities are carried out consistently. This process ensures operational stability, lasting success, and practical continuous improvement efforts.

4 **Promote a culture of continuous improvement:** This means emphasizing the development of a culture where the organization learns from experience, innovates, and adapts constantly.

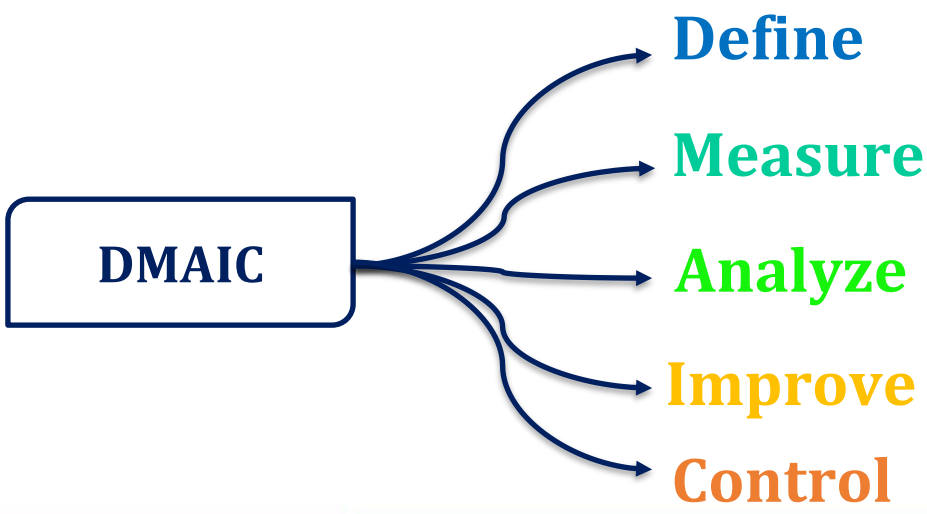
Iterative Loops of PDCA

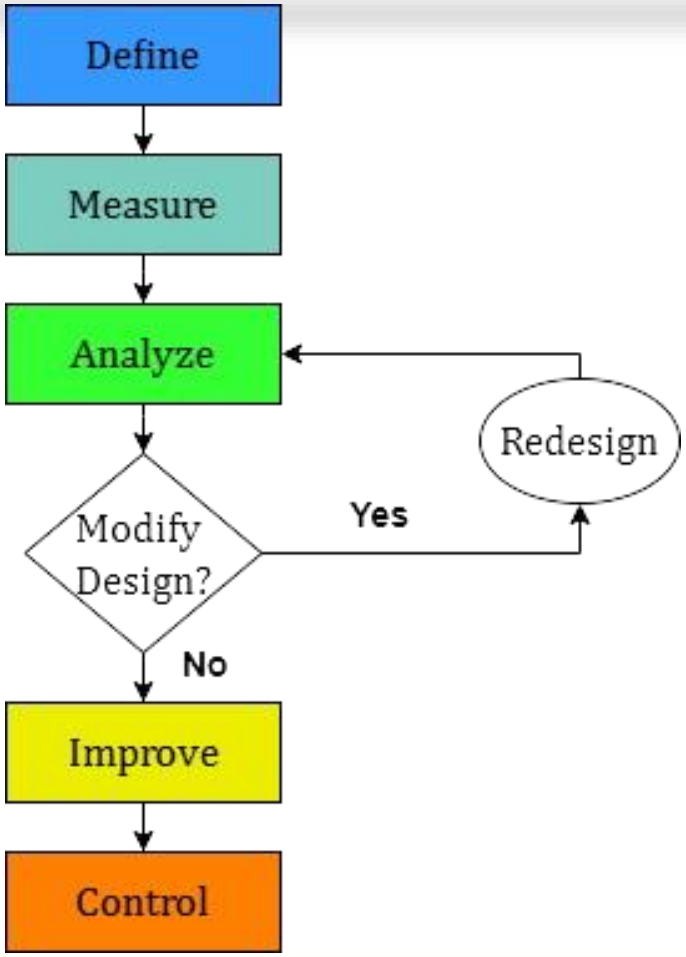
The PDCA (Plan Do Check Act) cycle is an **iterative process** because it involves a series of steps that are continually repeated and refined to achieve continuous improvement. The key to the iterative nature of the PDCA cycle is that, having completed the Act phase, one returns to the Plan phase, building on the knowledge and insights gained in the previous cycle.



Define Measure Analyze Improve Control (DMAIC)

DMAIC is the problem-solving approach that drives Lean Six Sigma. It's a five-phase method—Define, Measure, Analyze, Improve and Control—for improving existing process problems with unknown causes.





Define Measure Analyze Improve Control (DMAIC)



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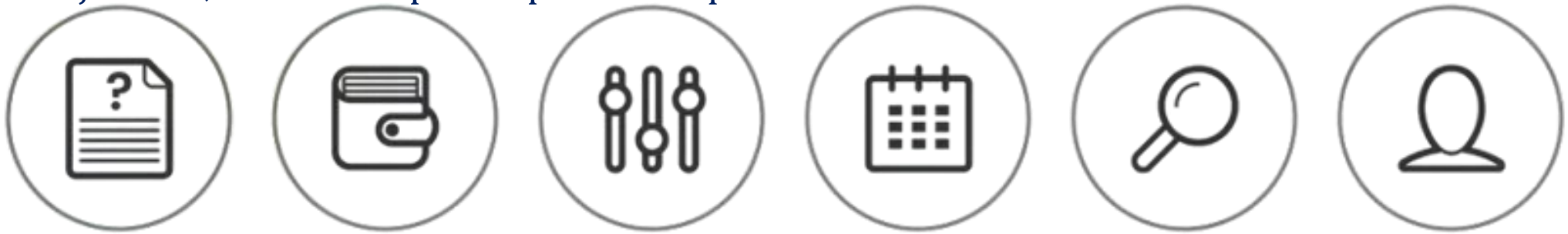
Define – DMAIC Phase 1

The Define phase is the first step in the Lean Six Sigma improvement process.

- In this phase, the project team creates a Project Charter, maps the process at a high level, and identifies the needs of customers related to the process.
- The team also conducts Process Walks (direct observations) and engages in discussions with individuals involved in the process to gain a deeper understanding of how it operates.
- Before proceeding to the Measure phase, the team ensures that the project focus is clearly defined and aligned with the organization's goals.

Define – DMAIC Phase 1

A **Project Charter** is a dynamic document for an improvement team that defines the existing problem, the objectives, and the scope of a process improvement initiative.



PROBLEM STATEMENT
The problem captured in the form of a measurement

BUSINESS CASE
The business reasons for doing the project

GOAL STATEMENT
The target of the process measurement

TIMELINE
When each project phase will be completed

SCOPE
What's in and what's out of the project

TEAM MEMBERS
The people who will participate in the project

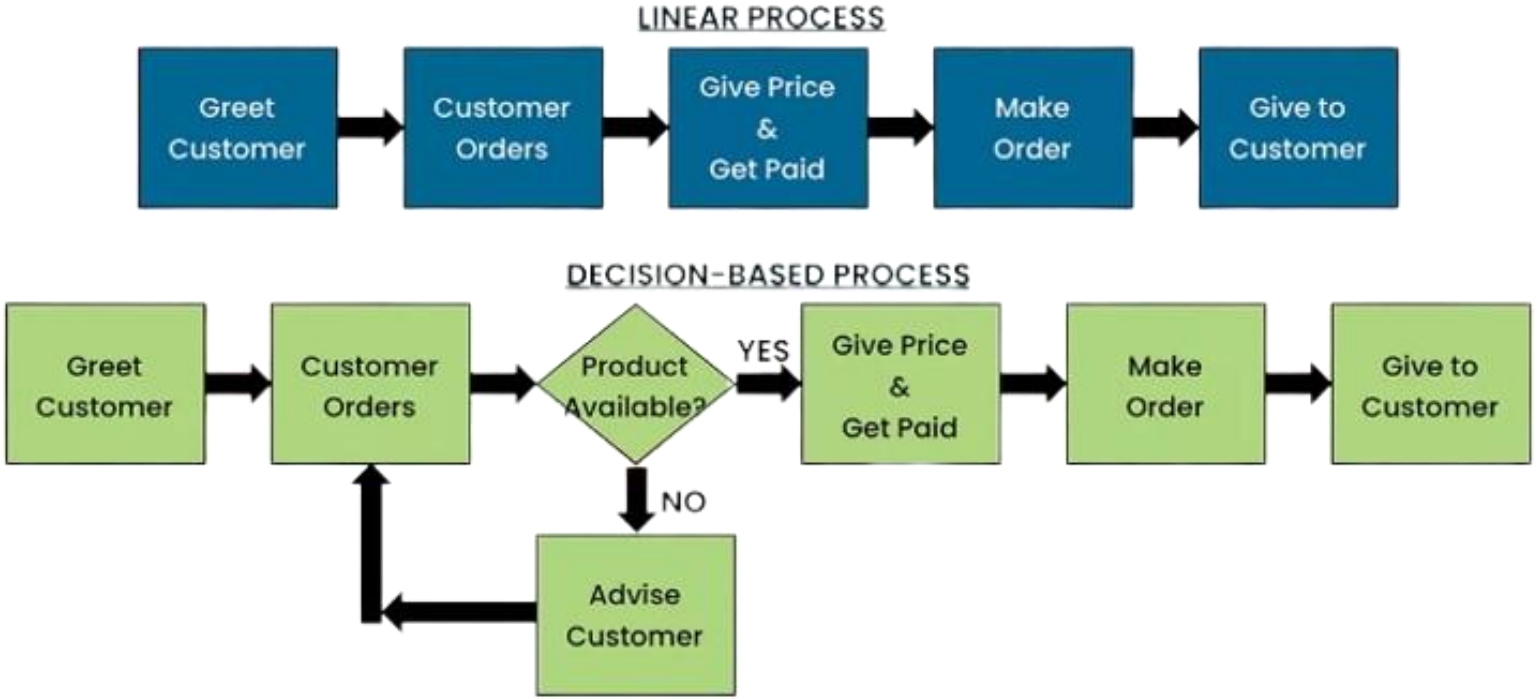
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Measure – DMAIC Phase 2

The **Measure** phase is crucial in a project as it provides key indicators of process health and helps identify where issues are occurring. As the team collects data, they focus on **process completion time** or the **quality of the output received by customers**.

Before proceeding to the **Analyze** phase, the team defines **measurement metrics** and assesses the **current performance** as a baseline for improvement.

A **Process Map** is a sequential diagram that illustrates the activities required to complete a process.



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Analyze – DMAIC Phase 3

The main focus of this phase is to ensure that hypotheses are tested and validated before implementing solutions. Once the analysis is complete and the root cause has been identified, the team can then proceed to the Improve phase.

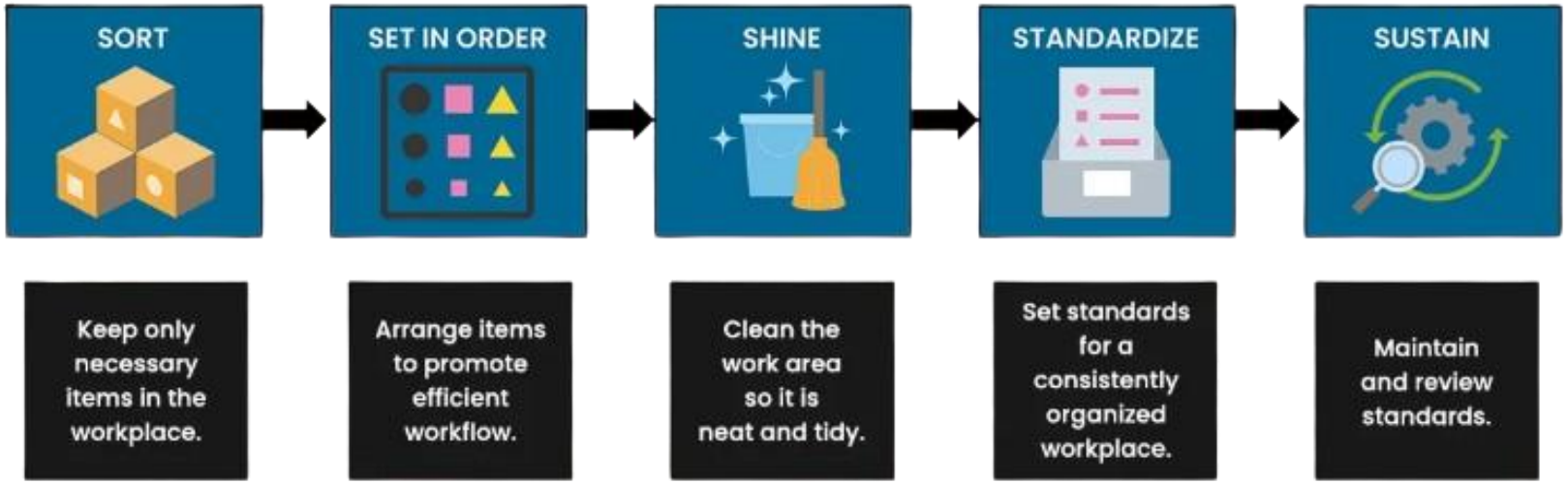
The 5 Whys is a simple analysis technique that goes beyond symptoms by repeatedly asking "why" until identifying the true root cause of an issue.



Improve – DMAIC Phase 4

The **Improve Phase** is where the team refines their countermeasure ideas, pilots process changes, implements solutions, and lastly, collects data to confirm there is measurable improvement. A structured improvement effort can lead to innovative and elegant changes that improve the baseline measure and, ultimately, the customer experience.

5S is a five-step organizing technique designed to create and maintain an intuitive workspace.

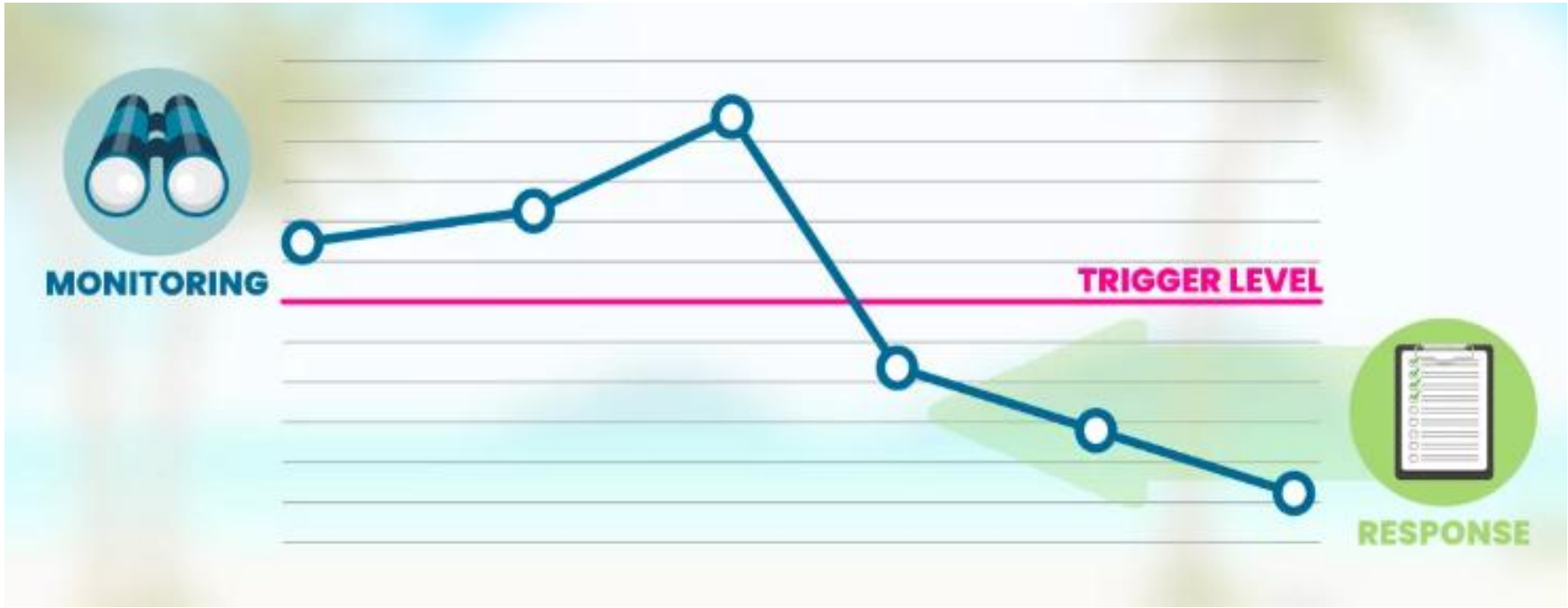


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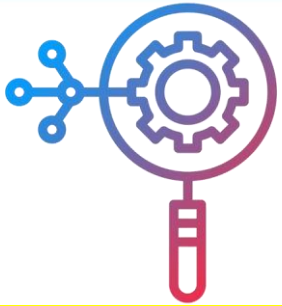
Control – DMAIC Phase 5

In the **Control Phase**, the team develops a Monitoring Plan to track the success of the updated process and crafts a Response Plan in case there is a dip in performance. Once in place, the Process Owner monitors and continually updates the current best method.

The Monitoring & Response Plan evaluates whether a process is functioning as expected and outlines actions to take if it is not.



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Root Cause Analysis

A **root cause** is defined as a factor that caused a nonconformance and should be permanently eliminated through process improvement. The root cause is the core issue—the highest-level cause—that sets in motion the entire cause-and-effect reaction that ultimately leads to the problem(s).
To help the team find answers to why a specific problem arises in your process, RCA (Root Cause Analysis) can be conducted in the following five steps:

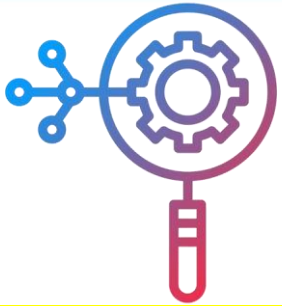
Step 1 – Define the Problem

- What problem is currently occurring?
- Describe the specific symptoms that indicate the existence of this problem!

Step 2 – Collect Data

- Do you have evidence proving that the problem actually exists?
- How long has this problem been present?
- What impact has been felt due to the problem?

At this stage, an in-depth analysis must be conducted before identifying the factors contributing to the problem. To ensure the Root Cause Analysis (RCA) is effective, gather representatives from all relevant departments (from experts to frontline staff) who understand the situation.

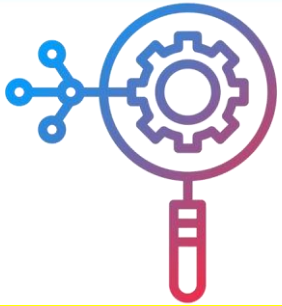


Root Cause Analysis

Step 3 – Identify Possible Causes

- Outline the sequence of events that led to the problem.
- Under what conditions does this problem occur?
- Are there any other issues that arise alongside or as a consequence of the main problem?

At this stage, identify as many potential causes as possible. In many cases, people tend to stop after identifying only one or two causal factors. However, one or two causes are often insufficient to determine the real root cause. RCA is not just about addressing surface-level problems but digging deeper to eliminate the root cause of the entire issue.



Root Cause Analysis

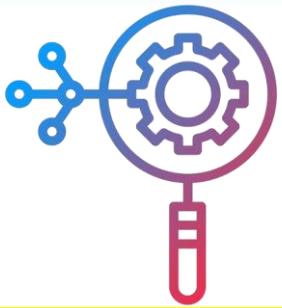
Use the following tools to help identify causal factors:

1. 5 Whys Analysis – Keep asking "Why?" until you reach the fundamental answer.
2. Drill Down – Break down the problem into smaller, more detailed parts to understand the bigger picture.
3. Appreciation – List the available facts and ask, "So what if this happens or doesn't happen?" to identify the most likely consequences of those facts.
4. Cause and Effect Diagram (Fishbone Diagram) – A visual tool that maps out all possible contributing factors to determine where the problem originated.

Step 4 – Identify the Root Cause(s)

- Why do these causal factors exist?
- What fundamental reason truly underlies the emergence of the problem?

Use the same tools from Step 3 to explore the root causes of each identified factor. These tools are designed to encourage deeper exploration at every level of cause and effect.



Root Cause Analysis

Step 4 – Identify the Root Cause(s)

- Why do these causal factors exist?
- What fundamental reason truly underlies the emergence of the problem?

Use the same tools from Step 3 to explore the root causes of each identified factor. These tools are designed to encourage deeper exploration at every level of cause and effect.

Step 5 – Propose and Implement Solutions

- What can be done to prevent the problem from recurring?
- How can the formulated solutions be implemented?
- Who will be responsible for implementing the solution?
- Are there any risks associated with implementing the solution?

Analyze your cause-effect identification process and assess whether changes are needed in other systems. It is also crucial to predict the potential effects of implementing the solution. By doing so, you can prevent or eliminate issues before they surface.



5 Whys Technique

The **5 Whys Problem Solving technique** is a simple process to follow to solve any problem by repeatedly asking the question "Why" (five times is a good rule of thumb), to peel away the layers of symptoms that can lead to the root cause of a problem. This strategy relates to the principle of systematic problem solving.

The 5 Whys Technique and Root Cause Analysis (RCA) are closely related because 5 Whys is one of the methods used in RCA to identify the root cause of a problem. RCA is a systematic approach to determining the fundamental cause of an issue, and 5 Whys serves as one of the techniques that support this process.



5 Whys Technique

Proper uses of the technique:

- To help identify the root cause of a problem
  - A root cause is the most basic reason, which if eliminated, would prevent reoccurrence.
- Provides a framework for a team to work through a more complex problem
- To solve problems as they occur

Improper uses of the technique:

- To emphasize the person or blame: turning the 5 Whys into the 5 Whos
- Making it a tedious, desk intensive project



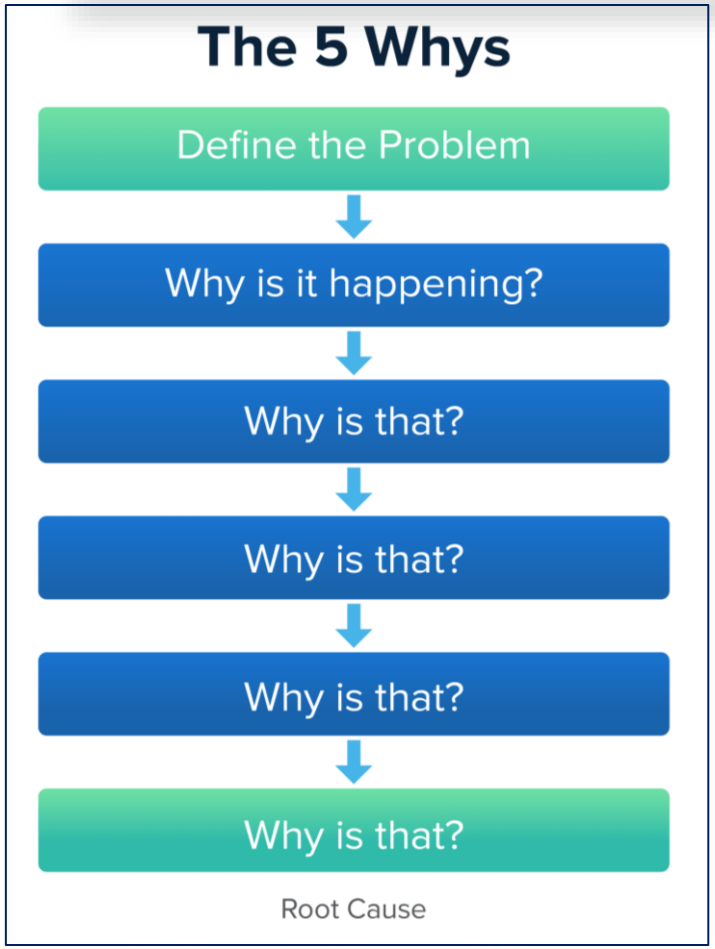
5 Whys Technique

Activity Procedure:

1. Write down the specific problem.
   - Writing the issue helps you formalize the problem and describe it completely.
   - It helps a team focus on the same problem.
   - Always describe the current condition.
   - Use data where possible.
     - Example: Overall customer complaints are up 50%.
2. Ask WHY the problem happens and write the answer down below the problem.
3. If the answer provided doesn't identify the root cause of the problem that you wrote in step 1, ask WHY again and write that answer down.
4. Loop back to step 3 until the team is in agreement that the problem's root cause is identified.



5 Whys Technique



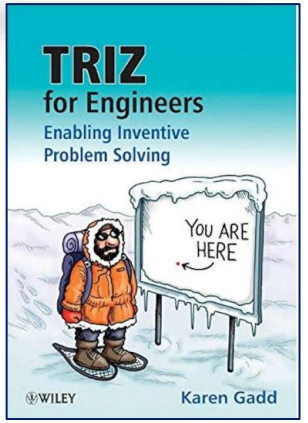
Case Example:

Problem Statement: Your company was unable to get the customer's product request to them on time.

1. Why were you unable to produce the product on time?
   - Because the equipment failed.
2. Why did the equipment fail?
   - Because the circuit board burned out.
3. Why did the circuit board burn out?
   - Because it overheated.
4. Why did it overheat?
   - Because the air filter wasn't changed.
5. Why wasn't the filter changed?
   - Because there was no afternoon preventative maintenance shift scheduled to change it.

Theory of Inventive Problem Solving (TRIZ)

Definition

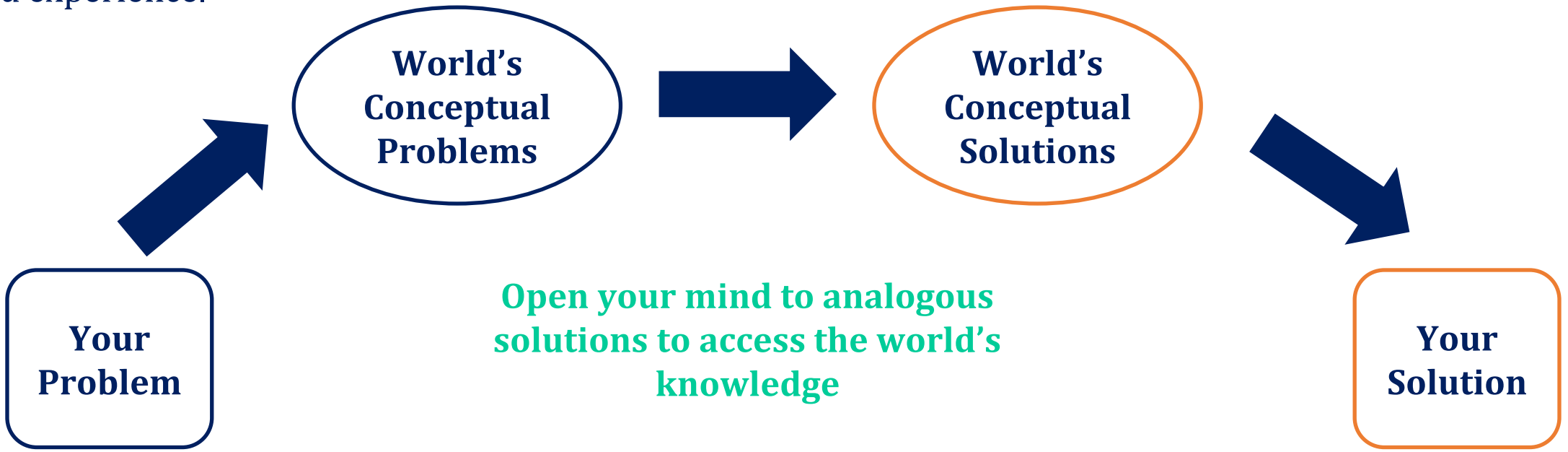


TRIZ is an engineering problem solving toolkit which successfully summarizes past solutions and successes to show us how to systematically solve future problems. TRIZ comes from Russia, initially and primarily the work of Genrich Altshuller, a great engineer and inventor, perhaps one of the greatest engineers of the twentieth century, whose work helps all other engineers.

TRIZ focuses our problem understanding to the particular, relevant problem model and then offers conceptual solutions to that model. Good engineers reduce wasted time with TRIZ as they head straight for the valid solutions and use their valuable time to define their problem accurately, find all the solutions to that problem and then develop those solutions.

Prism of TRIZ

The simplest TRIZ flowchart is the Prism of TRIZ which is a simple convergent /divergent tool for finding existing solutions, converging our real - life problems to a conceptual problem with conceptual answers and diverging the conceptual solutions back to real - life answers by adding back the relevant knowledge and experience.



Theory of Inventive Problem Solving (TRIZ)

Where Do We Start with TRIZ? Which Tools When?

There are simple stages which we can apply to any problem:

- ❑ Defining the Ideal for setting the direction of the problem solving and the scope for stakeholders.
- ❑ Problem Understanding – mapping the problem context in Time and Scale and capturing the relevant facts for both requirements and the system.
- ❑ Finding, creating and combining all the previous, spontaneous and known Solutions and Concepts. Filling up a ‘ Bad Solution ’ Park using everyone involved in the problem.
- ❑ Choosing/developing/transforming bad solutions into better solutions using TRIZ. Map all TRIZ transformed solutions in 9 boxes.

Thank you